Lab 6: Phasors - Part 2

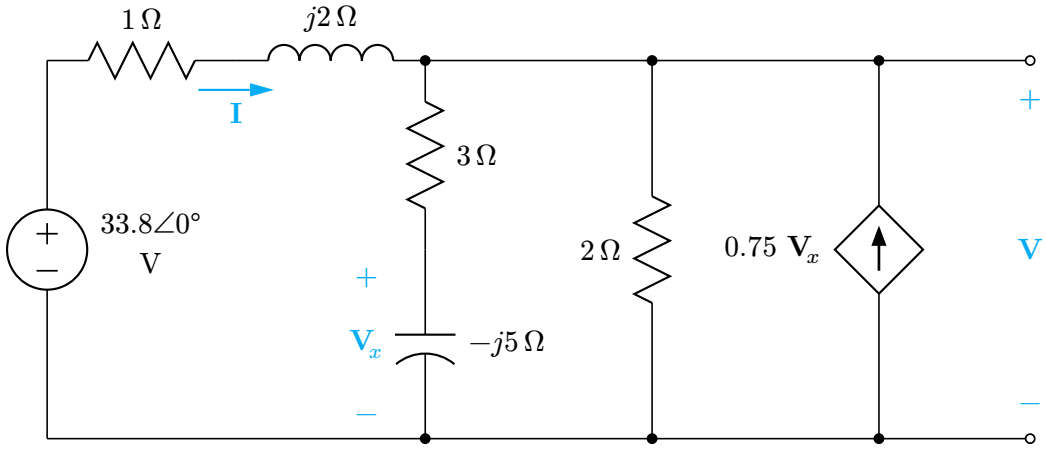
ECE2200L Section 04, Fall 2026

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No 1.

Step 1. Find the value of the phasor current $\mathbf{I}$ in the phasor-transformed circuit shown.



$$\mathbf{V}_x = \mathbf{V}\left(\frac{-j5}{3-j5}\right) = -\mathbf{V}\left(\frac{j5}{3-j5}\right)$$

$$\underbrace{\frac{\mathbf{V}-\mathbf{V}_s}{1+j2}}_{-\mathbf{I}} + \frac{\mathbf{V}}{3-j5} + \frac{\mathbf{V}}{2} - 0.75\mathbf{V}_x = 0$$

$$\frac{\mathbf{V}-33.8}{1+j2} + \frac{\mathbf{V}}{3-j5} + \frac{\mathbf{V}}{2} = -0.75\mathbf{V}\left(\frac{j5}{3-j5}\right)$$

$$\frac{\mathbf{V}}{1+j2} + \frac{\mathbf{V}}{3-j5} + \frac{\mathbf{V}}{2} + 0.75\mathbf{V}\left(\frac{j5}{3-j5}\right) = \frac{33.8}{1+j2}$$

$$\mathbf{V}\left[\frac{1}{1+j2} + \frac{1}{3-j5} + \frac{1}{2} + 0.75\left(\frac{j5}{3-j5}\right)\right] = \frac{33.8}{1+j2}$$

$$\mathbf{V} = 60.642\angle -81.66^\circ\,\text{V}$$

$$\mathbf{I} = -\frac{\mathbf{V}-\mathbf{V}_s}{1+j2}$$

$$\Rightarrow \boxed{\mathbf{I} = 29.069\angle 3.95^\circ\,\text{A}}$$

Step 2. Simulate the circuit to confirm your results in Step 1.

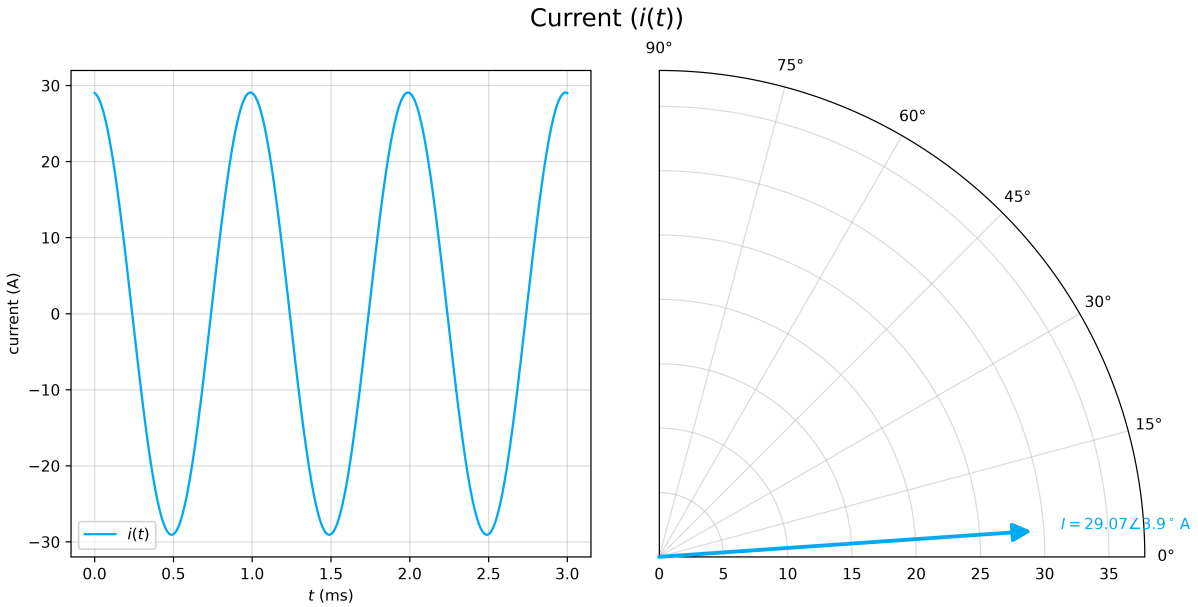
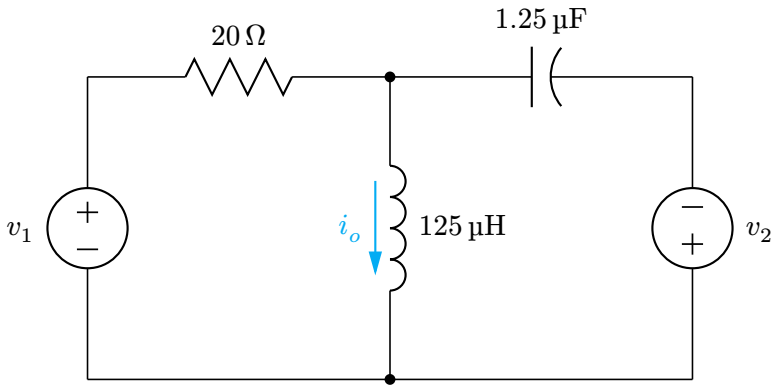


Figure 1: Time-domain current waveform illustrated at an assumed frequency of 1 kHz and corresponding phasor and corresponding phasor, $\mathbf{I} = 29.069\angle 35^\circ\,\mathrm{A}$.

No 2.

Step 1. Determine the time-domain expression for the steady-state value of $v_o(t)$ (voltage drop across the inductor) for the circuit shown below.



$$v_1(t) = 5\cos(80{,}000t)\ \mathrm{V}$$
$$v_2(t) = -2.5\cos(320{,}000t)\ \mathrm{V}$$

set $v_2 = 0$, replacing the ideal voltage source with a short circuit:

$$\omega = 80{,}000\ \mathrm{rad/s}$$
$$\mathbf{V}_1 = 5\angle 0^\circ\,\mathrm{V}, \quad Z_L = j10\,\Omega, \quad Z_C = -j10\,\Omega$$

notice:

$$Y_L + Y_C = -j\frac{1}{10} + j\frac{1}{10} = 0$$

this means the net current entering the inductor and capacitor is 0, meaning there is no voltage drop across the resistor:

$$\mathbf{V}_{o_1} = \mathbf{V} = 5\angle 0^\circ \,\mathrm{V}$$

now leave only v_2 on:

$$\omega = 320{,}000 \text{ rad/s}$$
$$\mathbf{V}_2 = -2.5\angle 0^\circ \,\mathrm{V}, \quad Z_L = j40\,\Omega, \quad Z_C = -j2.5\,\Omega$$

flip source and use voltage division

$$Z_{\parallel} = 20 \parallel Z_L = 16 + j8\,\Omega$$

$$\begin{aligned}\mathbf{V}_{o_2} &= (-\mathbf{V}_2)\frac{Z_{\parallel}}{Z_{\parallel} + Z_C} \\ &= 2.5\frac{16 + j8}{16 + j5.5} \\ &= 2.643\angle 7.59^\circ \,\mathrm{V}\end{aligned}$$

so,

$$v_{o_1}(t) = 5\cos(80{,}000t)\,\mathrm{V}$$
$$v_{o_2}(t) = 2.643\cos(320{,}000t + 7.59^\circ)\,\mathrm{V}$$

thus,

$$\boxed{v_o(t) = 5\cos(80{,}000t) + 2.643\cos(320{,}000t + 7.59^\circ)\,\mathrm{V}}$$

Step 2. Simulate the circuit to confirm your results in Step 1.

View simulation code

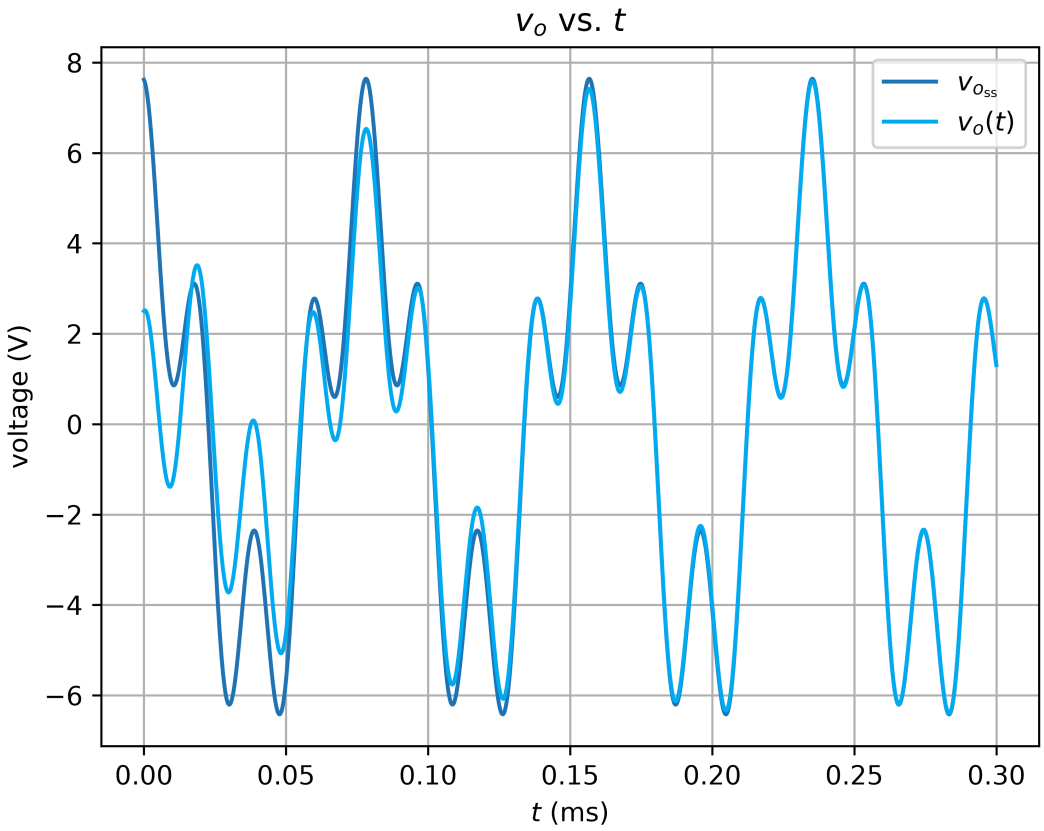
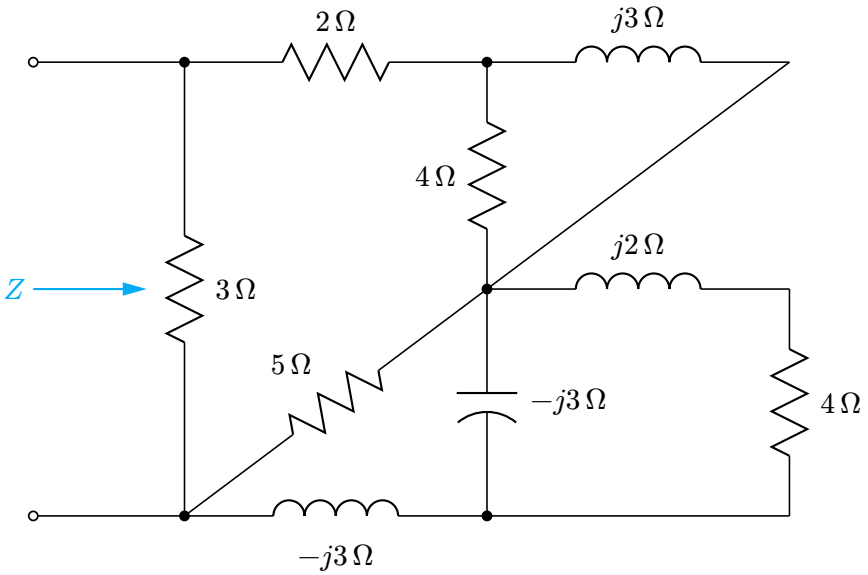


Figure 2: Comparison of the simulated inductor voltage and analytical steady-state response. The initial transient decays, and the simulated waveform approaches the analytical result.

No 3.

Step 1. Given the circuit below, compute the impedance Z



($-j3\,\Omega$ inductor likely a typo?)

$$\left(\underbrace{(4 \parallel j3)}_{Z_A} + \left(\left(\underbrace{(-j3 \parallel (4+j2)) - j3}_{Z_B}\right) \parallel 5\right) + 2\right) \parallel 3$$

$$Z_A = \frac{36}{25} + j\frac{48}{25}\,\Omega$$

$$Z_B = \frac{36}{17} - j\frac{93}{17}\,\Omega$$

then,

$$Z_C = (Z_B \parallel 5) + 2 = \frac{1313}{274} - j\frac{465}{274}\,\Omega$$

so,

$$Z = (Z_A + Z_C) \parallel 3$$

$$\boxed{Z = 2.0258 \angle 0.6654^\circ\,\Omega}$$

Step 2. Simulate the circuit to confirm your results in Step 1.

View simulation code

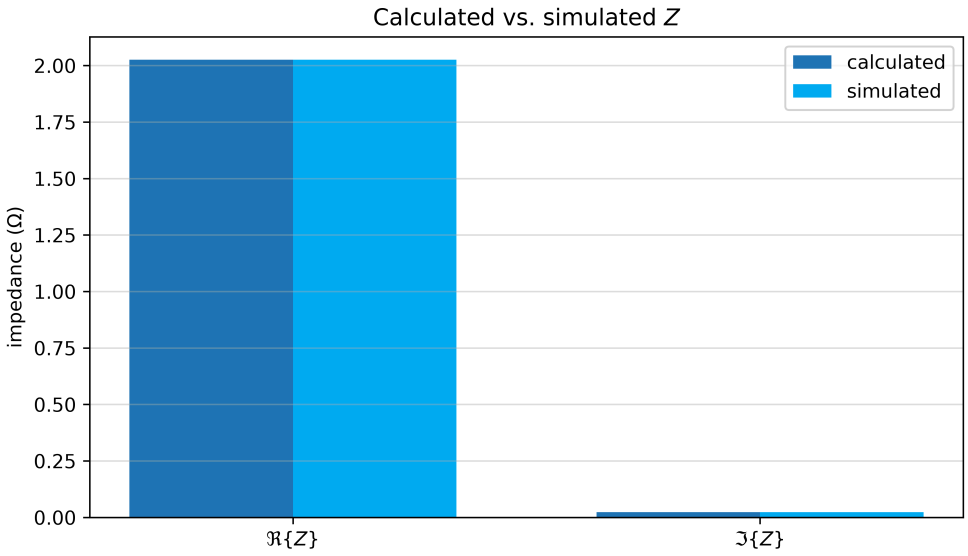


Figure 3: Comparison of the calculated and simulated real and imaginary components of the input impedance, showing agreement with $Z = 2.0258 + j0.6654^\circ\,\Omega$.